NRC7394 Evaluation Kit User Guide

(Smart Connection)

Ultra-low power & Long-range Wi-Fi

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NEWRACOM, Inc.

NRC7394 Evaluation Kit User Guide (Smart Connection)
Ultra-low power & Long-range Wi-Fi

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1 Overview

Smart Connection is a feature that automatically selects the optimal communication path in a network environment where multiple relay nodes exist. The selection process occurs during both the Initial Connection and subsequent Smart Connection operations.

2 Initial Connection

During the initial connection, the device operates without the Smart Connection feature enabled. Therefore, hop count or loop conditions are not considered — the device simply connects to the AP (or relay acting as SoftAP) with the strongest RSSI.

Example (see Figure 1):

- The Relay4 node boots up and attempts its first connection.
- It compares the RSSI values of nearby APs/Relays:
- Root (-50 dBm), Relay1 (-40 dBm), Relay2 (-30 dBm), Relay3 (-40 dBm).
- Relay4 connects to Relay2, which has the strongest signal (-30 dBm).

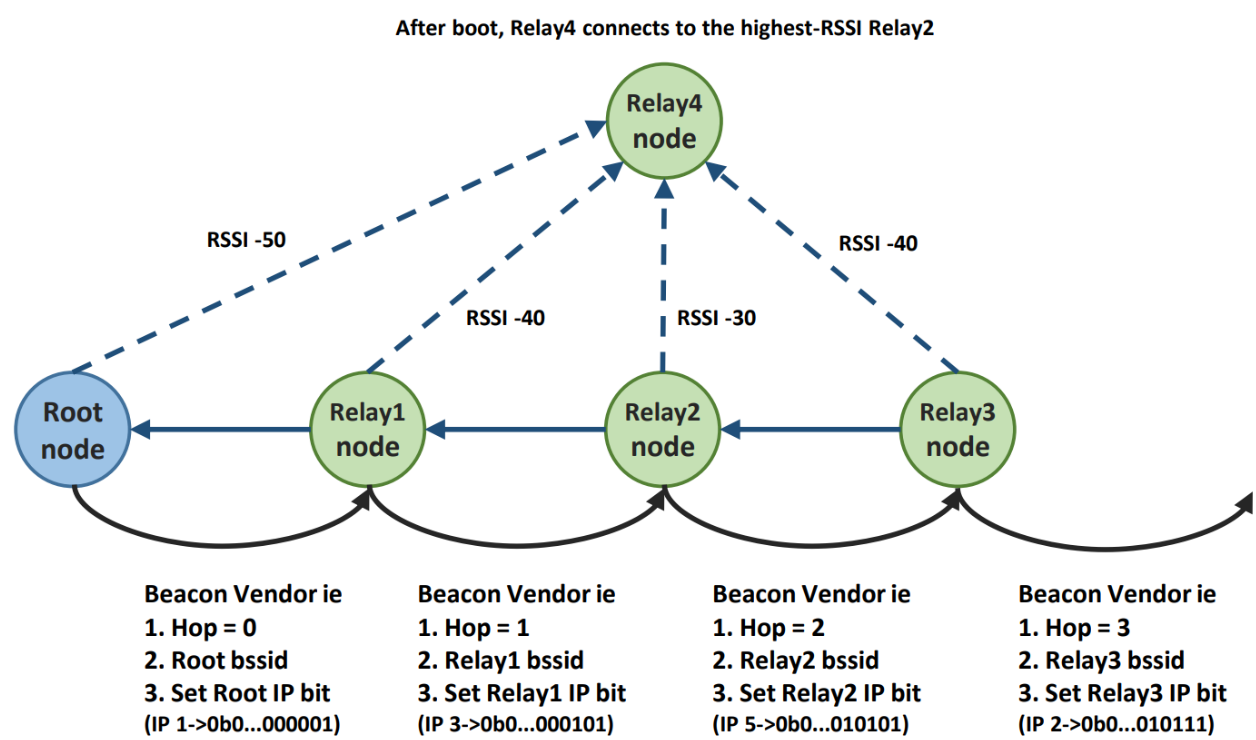


Figure 2.1 Multiple Relay connection networks

3 Smart Connection

3.1 Operation Overview (Vendor IE based)

The Smart Connection begins operation after the initial connection. Once the device (or relay AP) connects to the first AP, Smart Connection uses the Vendor IE information contained in beacons to determine the optimal path. Each AP/Relay broadcasts beacons containing Vendor IE fields, which include routing and loop-detection data.

Field	Description
Hop	Hop count of the AP/Relay AP - Root AP: Hop = 0 - Hop = -1 → path to Root disconnected
BSSID	MAC address of the AP/Relay
IP Bitmap	Accumulates the last bit of its IP address (256-bit size) e.g., IP X.X.X.1 → 0b0...0000001
RSSI	Not included in transmitted beacons; calculated by receiver firmware and stored in Vendor IE for Smart Connection decision logic

Table 3.1 Vendor IE fields for Smart Connection

3.2 Vendor IE Update Flow

Beacon Reception

The relay receives a beacon from its upstream node that contains a Vendor Information Element (IE) including the Hop count, BSSID, and IP Bitmap. At the time of reception, the receiver firmware measures the RSSI of the incoming beacon.

Vendor IE Update

The relay updates the received Vendor IE. The Hop value is incremented by one (except when the Hop is -1). The BSSID field is replaced with the relay's own BSSID, and the IP Bitmap is updated to include the bit corresponding to the relay's own IP address.

Beacon Transmission

The relay then broadcasts a beacon containing the updated Vendor IE. Other nearby devices and relays receive this beacon and use the included information to build their Smart Connection Scan List.

Example

- The Root node initially sends a beacon with Hop=0.
- Relay1 receives it and updates:
- Hop = 0 + 1 = 1
- BSSID = Relay1's BSSID
- IP Bitmap = adds its own IP bit

※ Relay Beacon Handling Rule

A relay can receive both upstream and downstream beacons, but only upstream ones are meaningful. Downstream beacons are ignored because their Vendor IE (Hop/IP Bitmap) indicates they are non-optimal paths.

3.3 Smart Connection Roaming Operation

The roaming decision in Smart Connection begins by checking whether the adjusted RSSI value is greater than or equal to the threshold sig_min (default = −60 dBm). Among the nodes satisfying this condition, the device prioritizes those with the lowest hop count. If multiple nodes have the same hop count, the scan list order is used to decide — the current BSS always appears first in the list, so it takes priority under identical conditions.

If there are no candidates even above −80 dBm, Smart Connection performs a last-resort roaming to the node with the strongest overall RSSI, regardless of hop count.

3.3.1 Periodic Scanning

The device performs scans every scan_interval (e.g., 20s) for scan_duration (e.g., 4s). It collects all beacons with the same SSID and builds a Smart Connection Scan List, including Hop, BSSID, and IP Bitmap information.

3.3.2 Roaming Decision

The relay determines its roaming behavior by selecting candidate nodes based on both RSSI and Hop count priority.

First, primary candidates are considered. A node qualifies as a primary candidate if its RSSI is greater than or equal to the minimum signal threshold (sig_min, default: –60 dBm) and its Hop count is less than or equal to 5. Among these, the relay selects the node with the lowest Hop count. If two or more nodes have the same Hop count, preference is given first to the current BSS, and then according to the scanning order.

The relay also keeps track of the strongest RSSI candidate—the node with the highest received signal strength overall—for last-resort roaming decisions.

- **Primary Candidates:**
    - RSSI ≥ sig_min (default: -60 dBm) and Hop ≤ 5
    - Select node with the lowest Hop count
    - If tied: current BSS first, then scan order.

- **Strongest RSSI Candidate**
    - Track node with the highest RSSI overall for last-resort selection.

Example

- (Case1) Primary Candidates Exist
    - When RSSI ≥ sig_min and multiple nodes have the same Hop value, the top node in the scan list (current BSS) is kept.

Node	Hop	Original RSSI	Adjusted RSSI	Decision
Current BSS	1	-69	-59	Stay
Node A	1	-62	-62	X
Node B	1	-58	-58	X
Node C	2	-50	-50	X

- (Case2) Strongest RSSI (Last Resort)
    - If all nodes < sig_min → roam to the node with the strongest RSSI (best_rssi).

Node	Hop	Original RSSI	Adjusted RSSI	Decision
Current BSS	2	-92	-82	X
Node A	3	-81	-81	Roaming
Node B	1	-85	-85	X
Node C	2	-88	-88	X

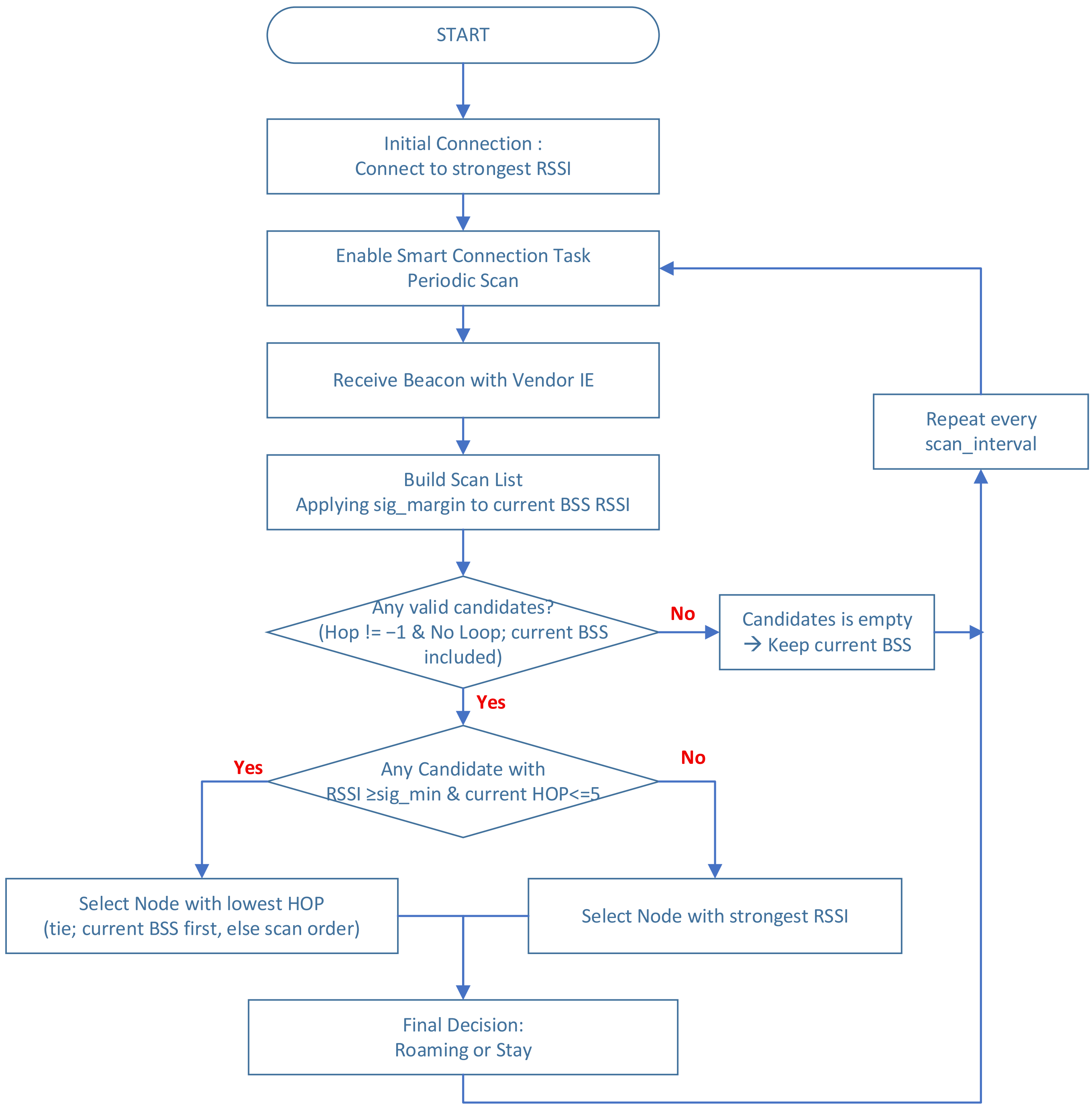


Figure 3.1 Smart Connection Roaming Decision Flow

3.4 Loop Detection Using IP Bitmap

The relay uses the IP Bitmap field in the Vendor IE to detect and prevent routing loops within the network. When the relay receives a beacon, the beacon's Vendor IE contains a cumulative IP Bitmap that represents all APs and relays the beacon has passed through.

- Relay Receives Beacon
    - Beacon's Vendor IE contains cumulative IP Bitmap representing passed APs/Relays.
    - (ex) 0b0...000101 → path already includes Root (bit 1) and Relay1 (bit 3).
- Check Own IP Bit
    - Relay verifies its own IP bit position within the bitmap.
- Loop Determination
    - If its bit is already set → loop detected → path is invalid.

Example

- Relay 1's assigned IP bit position is 3.
- Relay1 scans beacons and builds a Smart Connection list:

Node	IP	Hop	RSSI	Bitmap
Root	1	0	-40	0b...000001
Relay2	5	2	-30	0b...010101

- If Relay2's bitmap already has bit 3 set, it indicates that the communication path forms a loop. Therefore, Relay1 avoids roaming to Relay2 to prevent the loop.

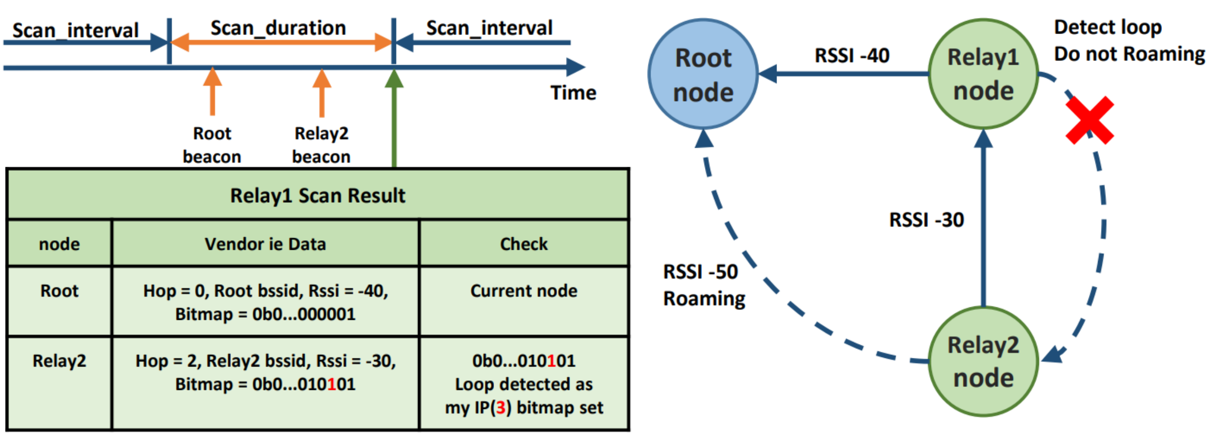
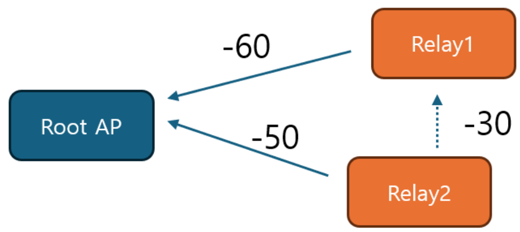


Figure 3.2 Smart Connection Roaming Example with Loop Detection

4 Example Scenarios

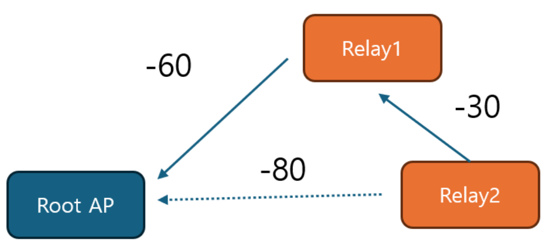
4.1 Scenario 1: Root → Relay1 → Relay2

- Initially, Relay2 connects to Relay1 (strongest signal).
- Later, Smart Connection re-evaluates:
    - Root AP and Relay2 RSSI ≥ -60 dBm
    - Hop count lower → Relay2 roams to Root AP.



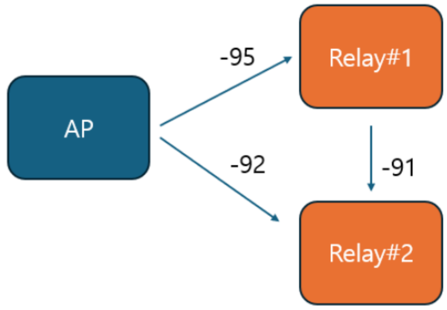
4.2 Scenario 2: Weak Root Signal

- Root AP RSSI = -80 dBm (below sig_min)
- Relay2 remains connected to Relay1 instead of Root.



4.3 Scenario 3: Weak Root Signal (RSSI < -80 dBm)

- Root AP and Relay #1 are already connected.
- When power is applied to Relay #2, it initially connects to Relay #1 due to the weak Root signal.
- Even after the smart connection feature operates later, Relay #2 remains connected to Relay #1 (similar to the second step in Scenario 2).


-95
Relay#1
AP
-92
-91
Relay#2

5 Revision History

Revision No	Date	Comments
Ver 1.0	10/21/2025	Initial version